

STAGE FIVE • THINKING, REASONING & MENTAL MODELS

FIRSTS

Thinking, Reasoning & Mental Models

SESSION 1

Foundational Mental Models

FIRSTS G1-G8

Mental Model Awareness

- You already use models, mostly unconsciously
- Naming a tool is step one to using it deliberately
- Start with a small, useful set, not dozens at once

Cause-and-Effect Mapping

- The real mechanism often stays invisible
- Don't stop at the first cause
- Correlation is not causation

Problem Breakdown Exercise

- Overwhelm → sequence of achievable steps
- Order matters as much as breaking down
- Often the real difference between finishing and stalling

Decision Matrix

- Structure beats “whichever option I considered last”
- Watch for weights that favor a predetermined answer
- Fewer criteria, chosen honestly

Root Cause Analysis

- Symptoms fixed without root causes resurface
- The first answer is often a symptom
- Honest diagnosis, not self-blame

Critical Thinking Drill

- Every claim rests on unstated assumptions
- Apply equal scrutiny to claims you like
- Practice makes it a reflex, not an occasional effort

Hypothesis Testing

- Untested assumptions shape decisions constantly
- Small experiments replace guessing with evidence
- A real test can actually disprove your prediction

Scenario Planning

- Best case, worst case, most likely — map all three
- Worst-case-only breeds anxiety, not preparation
- Most-likely is often the most useful, least exciting

SESSION 2

Applied Reasoning & Reflection

FIRSTS G9-G15

Cognitive Bias Audit

- Everyone has biases; few think their own reasoning is biased
- Naming a recurring bias makes it catchable
- An ongoing watch, not a one-time label

Systems Thinking Exercise

- Isolated failures are often systemic symptoms
- Fixing the symptom alone lets it resurface elsewhere
- Real systems usually have more than one loop

Feedback Integration

- Unintegrated feedback = no feedback
- Value appears only once it changes your next action
- Repeated feedback is a signal, not nagging

Strategic Prioritization

- Easy-but-low-impact crowds out hard-but-high-impact
- Not everything can be top priority
- Urgency $\neq$ importance

Mental Model Application

- Knowing about a model $\neq$ applying it under real conditions
- Write the reasoning out, don't just think it loosely
- Ties together everything from Session 1

Pattern Recognition Drill

- One data point rarely tells you much
- Three or four instances reveal a genuine pattern
- Don't overgeneralize from coincidence either

Reflection on Decision-Making

- Without reflection, mistakes repeat in isolation
- Include the decisions that went poorly, honestly
- Closes the loop on the G category

SESSION 3

Advanced Frameworks I

FIRSTS H16-H23

SWOT Thinking Exercise

- Works for personal decisions, not just business
- Internal + external factors, together
- Specific to the real decision, not generic

First-Principles Analysis

- Precedent is efficient but can trap you
- Strip assumptions down to what's truly required
- Not the same as just being contrarian

Inversion Thinking Drill

- Failure is often easier to see than success
- Use that asymmetry deliberately
- Translate the failure list into real prevention

Opportunity Cost Exercise

- Anything looks reasonable evaluated alone
- Name the real next-best alternative
- “Vs. doing nothing” understates the real cost

80/20 Analysis

- Effort and results are rarely evenly distributed
- A small subset often drives most real results
- A pattern to investigate, not an exact ratio

Systems Thinking Application

- Personal-scale skill → real team impact
- Most team problems are systemic, not individual
- Don't default back to blaming people when it gets hard

Decision Tree Exercise

- Multi-step decisions overload working memory
- A visual tree externalizes the structure
- Should simplify, not overwhelm further

Scenario Analysis

- Scales G8's scenario planning to real complexity
- Focus on the 2-3 variables driving real uncertainty
- Still needs a clear, actionable plan per scenario

SESSION 4

Advanced Frameworks II

FIRSTS H24-H30

Checklists & SOPs Creation

- Memory alone accumulates small, repeated errors
- A checklist catches what memory misses
- Too long to use defeats its own purpose

Root Cause Framework

- A named method beats reinventing the dig each time
- Extends G5 into a formal, repeatable process
- Five whys is a guideline, not a rigid rule

Feedback Loop Exercise

- Try → observe → adjust, deliberately
- Getting it right first try is the exception
- Record and review each cycle, or it's just repetition

Comparative Analysis

- Uneven comparisons (cost vs. gut feeling) are unreliable
- Same explicit criteria, every option
- Extends G4's Decision Matrix

Mental Model Combination

- Real problems rarely fit one clean model
- Combining models can reveal what neither shows alone
- Called for by the problem, not performed to impress

Hypothesis-Driven Approach

- Big plans on untested assumptions risk big waste
- Small test first, extends G7 to higher stakes
- “Obviously true” is exactly what's worth testing

Strategic Mapping

- Connect near-term actions to the longer-term goal
- The capstone — combines prioritization, systems, scenarios
- Revisit it; an outdated map still gets used anyway

Stage Five, Complete

One mental model you'll keep using deliberately, named out loud.